USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 165Lgw (35628R), Sustainability Science in the City

Syllabus

Units: 4

Term Day Time: Spring 2026
Lecture: MWF 11:00-11:50am
Location: THH 102

Instructor: Bita Minaravesh, Ph.D., MPH
Office: AHF B55D
Regular Office Hours: Mondays, 12:00 – 1:00 p.m. and
Tuesdays, 11:00 – 12:00 p.m.
Contact Info: minarave@usc.edu

Lab Times & Location:
Mondays 2-3:50 p.m. (ZHS 360);
Mondays 4-5:50 p.m. (THH B9);
Wednesdays 12:00-1:50 p.m. (DMC 155);
Wednesdays 4:00-5:50 p.m. (SOS B41);
Thursdays 10:00-11:50 a.m. (KDC 236);
Fridays 12:00-1:50 p.m. (THH 117)

Laboratory Co-Instructor: TBD
Office: AHF B55
Regular Office Hours: TBD
Contact Info: TBD

Library Help: Andy Rutkowski
Office: LIPA B40-A
Office Hours: Available by appointment via email.
Contact Info: arutkows@usc.edu, see contacts module on Brightspace for Zoom Room

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu

Course Scope and Purpose

Sustainability is among the most pressing scientific and social challenges of our time. Typically defined as utilizing natural resources to create a high quality of life for *future* as well as current generations, the idea of sustainability has provided a strong orientation towards a long-term re-thinking of the human role in and domination of ecosystems. Yet, despite the emergence of a sustainability policy discourse in the late 1980s, global climate change, ocean degradation, deforestation, habitat loss, and species endangerment remain a serious challenge to ecosystem services, health, and well-being for present and future generations.

In response to such ongoing challenges, the field of *sustainability science* emerged in the late 1990s. It is a multidisciplinary collection of social, physical, and life sciences that work to understand the complex coupling of human and natural systems across global, national, regional, and local scales. Envisioning a sustainable future requires deep understanding and reconsideration of the human role in natural systems. Thus, policymakers rely upon various forms of scientific knowledge and the scientific method itself to understand how to re-chart the human journey towards sustainability.

This course is a Social Analysis (Category C) and a Global Perspectives: Equity in a Diverse World (Category G) course in USC's General Education program. This course is also included at the start of USC's GIS and Sustainability Science (GISS) minor. In this course, you will learn how social and ethical theories of sustainability relate to the emergence of sustainability science and how theory and empirical work are mutually constitutive. You will learn why the social and natural sciences and their methods are important to policies and planning for sustainable cities, and how data are used from the social and natural sciences to assess the validity of arguments about reshaping cities for sustainability.

In 2008, an important global threshold was reached, with over 50% of people living in cities. According to UN forecasts, by 2050 70% of the Earth's growing population will be living in urban areas. The rapid growth of cities across the world results from a common undercurrent of global political and economic forces that rests on a history of colonialism. We will strive to understand how these forces might be reshaped to create sustainable forms of urban development. We will explore land use, population, consumption, urban ecology and infrastructural technologies (e.g., energy, transportation, etc.).

In a series of laboratory experiences and linked writing exercises, you will learn how to articulate spatial relationships, the analytical approaches and methods used to understand them, and their societal implications. For example, one focus of sustainability science is improving our understanding of how the Earth's land cover and land use is changing, and what it means for people and places. In the laboratory, you will learn geographic information systems (GIS) and satellite imagery, are used for measuring land use/land cover change and how observed land use/land cover changes are linked to principles of urban form and urban economics. In related writing assignments, you will consider land use/land cover in the context of social well-being, equity, sustainable livelihoods, and land use politics and regulation.

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Learning Outcomes

Upon successfully completing this course, students should be able to:

- Describe the sustainability problematic and the problem of urban sustainability as a coupled human and natural system;
- Classify both natural and social systems in cities and explain how they are interwoven;
- Understand how data collection, non-experimental research design, and computational modeling are used to analyze urban sustainability problems and solutions;
- Use and interpret spatial data in a professional geographic information system (GIS) to analyze issues of urban sustainability;
- Communicate the science and policy of urban sustainability by leveraging digital resources and applying basic cartographic principles;
- Articulate alternative pathways toward sustainable cities and evaluate scientific claims related to such alternatives;
- Articulate concepts of environmental justice both within the U.S. and globally, including the idea of sustainable citizenship;
- Assess how applications of scientific methodologies influence or are influenced by debates over urban sustainability.

Students may vary in their competency levels on these abilities. You can expect to acquire these abilities only if you attend classes regularly, do the course readings, and carefully read and complete assignments.

Prerequisite(s): None
Co-Requisite(s): None

Class Conduct

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the Title IX Coordinator. The Title IX Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The Title IX office is also responsible for coordinating supportive measures for transgender and nonbinary students such as faculty notifications, and more. If you need supportive accommodations, you may contact the Title IX Coordinator directly (titleix@usc.edu or 213-821-8298) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours)

Diversity and Inclusion – I intend that students from all backgrounds and perspectives will be well served by this course, that students' learning needs will be addressed both in and out of

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class, and that the diversity that students bring to this class will be viewed as a resource, strength, and benefit. I aim to present materials and activities that are respectful to everyone, and you are also expected to respect others regardless of their race, ethnicity, gender identity and expressions, cultural beliefs, religion, sexual orientation, national origin, age, abilities, ideas and perspectives, or socioeconomic status. Your suggestions are encouraged and appreciated. Feel free to let me know ways to improve the effectiveness of the course for yourself or for other students.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class is forbidden without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion, and thus infringe on the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposes other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

Required Readings and Supplementary Materials

Please acquire the texts listed below. Both are available at the USC bookstore. All other supplementary reading listed in the syllabus is available in the related module on Brightspace. The reading is listed by author and chapter number and the course schedule below, and you should aim to read it before coming to lecture on the day it listed.

The required textbooks for this course are:

- Hagen, Bjoern and K. David Pijawka, eds. 2024. *Sustainability for the 21st Century: Pathways, Programs, and Policies,* 4th Edition. Dubuque, IA: Kendall Hunt. This book, it is available online in the My Materials tab on the Brightspace homepage with a code you can obtain at the USC Bookstore.
- Wheeler, Stephen M. 2013. *Planning for Sustainability: Creating Livable, Equitable, and Ecological Communities*, 2nd Edition. New York: Routledge. This book, it is available online in the My Materials tab on the Brightspace homepage.

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Supplementary readings for this course, located within the related course modules on Brightspace, include:

Chapters selected from the following book:

- Sims, Kearrin, Nicola Banks, Susan Engle, Paul Hodge, Jonathan Makuwira, Naohiro Nakamura, Jonathan Rigg, Albert Salamanca, and Pichamon Yeophantong, eds. 2022. *The Routledge Handbook of Global Development*. New York, NY: Routledge. This entire book is available as an electronic book at the USC library. We will read carefully selected chapters as listed by chapter number in the course schedule. The complete citation for each selected chapter is listed below:
    - Andriesse, Edo and Kristian Saugin. 2022. "Spatial Inequality and Development" pp. 351-361 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 30)
    - Bello, Walden. 2022. "Deglobalization" pp. 20-32 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 2)
    - Batista, Tanja and Ronald Skelton. 2022. "International and Internal Migration" pp. 533-544 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 45)
    - Gregoire, Etienne Roy and Pascale Hatcher. 2022. "Global Extractivism and Inequality." pp. 341-350 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 29)
    - Krishnan, Aarti. 2022. "Global Value Chains and Development" pp. 516-532 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 44)
    - La Viña, Antonio G.M. and Jameela Joy M. Reyes. 2022. "Anthropocene, Capitalocene, and Climate Change" pp. 158-169 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 14)
    - Marks, Danny. "Transboundary Governance Failures and Southeast Asia's Plastic Pollution" pp. 280-289 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 24)
    - Mitilin, Diana, et al. 2022. "Citizenship, Rights, and Global Development" pp. 493-504 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 42)
    - Rigg, Jonathan and Kearrin Sims. 2022. "Poverty: No Meeting of the Minds" pp. 311-325 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 27)
    - Veltmeyer, Henry 2022. "Extractivism" pp. 197-206 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 17)

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- Vera-Sanso, Penny. 2022."Ageing and Development" pp. 576-586 in K. Sims et al. (eds) *The Routledge Handbook of Global Development*. New York, NY: Routledge. (Chapter 49)

The following articles and book chapters, listed by author in the course schedule:

- Anjali, Mahendra and Victoria Beard. 2018. "Achieving Sustainable Cities by Focusing on the Urban Underserved: An Action Agenda for the Global South" pp. 411-416 in T. Elmquist et al. (eds.) *Urban Planet: Knowledge Towards Sustainable Cities*. New York, NY: Cambridge Press.
- Bullen, Anna and Mark Whitehead. 2005. "Negotiating the Networks of Space, Time, and Substance: A Geographical Perspective on Sustainable Citizen." *Citizenship Studies* 9, No. 5: 499-516.
- Cook, E. 2020. "Urban Ecology, Green Networks, and Ecological Design" pp. 205-222 in B. Hagen. and K.D. Pijawka (eds.) *Sustainability for the 21st Century* (3rd Edition). Dubuque, IA: Kendall Hunt.
- Cronon, William. 1996. "The Trouble with Wilderness: Or, Getting Back to the Wrong Nature." *Environmental History* 1, no. 1 (January): 7-28
- Goldstein, Noah J., Robert B. Cialdini, and Vladas Griskevicus. 2008. "A Room with a Viewpoint: Using Social Norms to Motivate Conservation in Hotels." *Journal of Consumer Research* 35: 472-482.
- Guhathakurta, Subhrajit. 2017. "Examining Urban Sustainability through Urban Models" pp. 111-126 in B. Hagen and K. D. Pijawka (eds.) *Sustainability for the 21st Century* (2nd Edition). Dubuque, IA: Kendall Hunt.
- Maniates, Michael. 2002. "Individualization: Plant a Tree, Buy a Bike, Save the World?" In *Confronting Consumption*, edited by Thomas Princen, Michael Maniates and Ken Conca, 43-66. Cambridge, MA: MIT Press.
- McAslan, Devon. 2015. Assessing Urban Sustainability: Using Indicators to Measure Progress" In *Sustainability for the 21st Century* (1st Edition), edited by K. David Pijawka, 235-258. Dubuque, IA: Kendall Hunt.
- Princen, Thomas. 2002. "Consumption and its Externalities: Where Economy Meets Ecology." In *Confronting Consumption*, edited by Thomas Princen, Michael Maniates and Ken Conca, 23-42. Cambridge, MA: MIT Press.
- Pulido, Laura, Steve Sidawi, and Robert O. Vos. 1996. "An Archeology of Environmental Racism in Los Angeles." *Urban Geography* 17, no. 5: 419-439.
- Simkin, Rohan D., Karen C. Seto, Robert I. McDonald, and Walter Jetz. 2022. "Biodiversity Impacts and Conservation Implications of Urban Land Expansion Projected to 2050." *Proceedings of the National Academy of Science* 119, no. 12: e2117297119.
- Webster, Douglas, Feifei Zhang, and Jianming Cai. 2017 "China's Pursuit: Smart Sustainable Urban Environments" pp. 307-331 in B. Hagen and K. D. Pijawka (eds.) *Sustainability for the 21st Century* (2nd Edition). Dubuque, IA: Kendall Hunt.

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- Vos, Robert O. 2007. Defining Sustainability: a Conceptual Orientation." *Journal of Chemical Technology and Biotechnology* 82: 334-339.
- Wolch, Jennifer, Jason Byrne, and Joshua P. Newell. 2014. "Urban Green Space, Public Health, and Environmental Justice: the Challenge of Making Cities 'Just Green Enough'." *Landscape and Urban Planning* 125: 234-244.

Description and Assessment of Assignments

Students must attend all regularly scheduled lectures/in-class exercises, participate in labs, write in the reading journal discussion forum, deliver a city sustainability profile, sit for mid-term and final examinations, and produce a final project called a "story map."

Labs

In addition to the lectures and in-class exercises, there is a set of 10 labs and related deliverables across the semester. These laboratory experiences are designed to introduce you to the tools of spatial and social analysis as well as to give you practical experience in implementing these tools to explore various problems within the framework of the scientific method. These assignments are linked to the lectures, but do not duplicate the lecture experience. You must register for one laboratory session in addition to registering for the lectures.

Absences from lab sessions must be requested *in advance* by sending an email to the laboratory instructor for your lab section. Excused absences from labs will be granted only for valid reasons; please notify us of the reason for your absence in your email.

Reading Journal Discussion Forum

At the start of the semester and again after the midterm, students will participate in a reading journal discussion forum. Here you will find an article from major press outlets (e.g., CityLab, Wall Street Journal, New York Times, and Los Angeles Times) on an issue of sustainable cities that relates to one of our assigned course readings. You will post a short summary of the article and explain how it relates to the reading. You will also read and briefly comment on posts by two of your classmates.

Story Map

The final project in this course is a story map, which will present data on social and natural systems on a specific topic related to one of our course modules. It will be planned and may mostly be accomplished during lab sessions as indicated on the lab schedule. Story maps tell about places, issues, and trends by enriching digital maps with content like various kinds of graphs, text, photographs, video, and audio. The underlying data often depict the coupling of social and natural systems. These may be things like wetland areas, land cover, and census data, and may also include live data streams such as temperature, precipitation, and traffic. They often present scientific data and analysis, but they are mainly designed for the public and do not require their users to have special knowledge or skills in GIS.

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Story maps are increasingly in use in sustainability science and are an important tool to describe the challenges of sustainable cities and pathways toward sustainability. For example, you can see an interactive story map that describes land use footprints of megacities here: http://storymaps.esri.com/stories/2014/growth-of-cities/. This story map was created as part of the Smithsonian's series on *Living in the Anthropocene: The Age of Humans*.

In this course, you will create a story map that integrates data on social and natural systems around one of our course's learning modules. Your story map will integrate scientific data like the example above but may be focused at either local or global scales. Some story maps are simply a montage of geotagged photographs. Your story map will be much more than this. It may have photos for context, but it must be primarily an analytical report that includes writing in pop-up windows and sidebars. It will use visualization of data or models to communicate underlying analysis.

Exams and Other Policies

The midterm and the final exam are closed book. The mid-term and final exams will include content learned in course readings, lectures, in-class exercises, laboratory sessions, and assignments up until the date of each exam. Except for *documented* illness, emergencies, or official USC conflicts, **no make-up opportunities will be offered for missed exams or labs,** so mark the appropriate dates on your calendars! If you have a legitimate excuse, speak with the instructor as soon as possible to arrange a make-up. Also, note that unless excused **in advance** for *documented* illness, emergencies, or official USC conflicts, there is **no credit for late assignments.**

Grading Breakdown

The following table shows the breakdown of the assignments and their weight in the final grade. The emphasis is on regularly completing several short assignments as well as solid performance on examinations and the final project. Assignments must be submitted as noted, typically on the appropriate Brightspace site.

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Assessment	Number	Total Points (% of Grade)
Laboratory Reports *Note: Lab reports are on the course schedule at the start of each week they are assigned. Submit all Lab Reports via the Brightspace for your laboratory section before the next week's lab session starts.*	6	24
Reading Journal Discussion Forum (Submit on the Lecture Brightspace)	2	12
Story Map Sessions & Deliverables	5	15
Midterm Exam (In class closed book)	1	11
Final Exam (Closed book at scheduled exam time)	1	20
Final Project: Story Map (Submit URL to Lab Section Brightspace and give presentation at final lab session)	1	18
Totals	16	100

Grading Scale

Assignments in this and other SSCI courses, are graded on a percentage of the total points available for a given assignment. The grading scaled corresponds to letter grades where A is exemplary, B is very good, C is satisfactory, D is unsatisfactory, and F is failing. Final letter grades are assigned according to the total points earned in the course using standard rounding rules. The grading scale is as follows:

A	> 93 points		B-	80-82 points		D+	67-69 points
A-	90-92 points		C+	77-79 points		D	63-66 points
B+	87-89 points		C	73-76 points		D-	60-62 points
B	83-86 points		C-	70-72 points		F	<60 points

Grading Timeline

My goal is to provide grading and feedback on each course assignment with sufficient time for you to take account of feedback before the next related assignment is due. In general, you might expect lab reports to be graded within 1-2 weeks as these assignments occur regularly. Other writing and presentation assignments may be returned within about 3 weeks.

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Learning Experience Evaluations

Please note Learning Experience Evaluations for the course take place at the end of the semester and are facilitated by the University. These evaluations provide an important review of student experiences in the course.

Schedule

Modules

1	Theories and Key Concepts
2	Environmental Footprints
3	Environmental Governance
4	Greening the Urban Economy and Urban Metabolism
5	Reclaiming Urban Nature
6	Climate Change: Mitigation and Adaptation
7	Environmental Justice (EJ): Community-Based Sustainability Science
8	Perspectives on Consumption
9	Indicators, Tools, and Sustainable Citizenship

M	Date	Topics	Readings	Deliverables/Due Dates
1	**Week 1** 1/12	Introduction to the Course		*No Lab Meeting.*
	1/14	The Urban Sustainability Problematic	Hagen and Pijawka Ch. 1 Wheeler Ch. 1 & 4 Sims et al. Ch. 14	
	1/16	Competing Definitions of Sustainability and Sustainable Cities	Hagen and Pijawka Ch. 2 Wheeler Ch. 2 Vos, R. O. 2007. "Defining sustainability: a conceptual orientation." *Perspective in Journal of Chemical Technology and Biotechnology* 82: 334-339.	

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	Week 2 1/19	*MLK Jr. Holiday (No class meeting)*		*No Lab Meeting.*
	1/21	The Global Context of Sustainable Cities (1)	Hagen and Pijawka Ch. 3 and 6 Wheeler Ch. 19 Sims et al. Ch. 2 & 17 Webster et al. 2017 "China's Pursuit: Smart Sustainable Urban Environments" pp. 307-331 in B. Hagen and K. D. Pijawka (eds.) *Sustainability for the 21st Century* (2nd Edition). Dubuque, IA: Kendall Hunt.	
	1/23	The Global Context of Sustainable Cities (2)	Sims et al. Ch. 27, 29, & 30 Anjali, M. and V. Beard. 2018. "Achieving Sustainable Cities by Focusing on the Urban Underserved: An Action Agenda for the Global South" pp. 411-416 in T. Elmquist et al. (eds.) *Urban Planet: Knowledge Towards Sustainable Cities*. New York, NY: Cambridge Press.	
	Week 3 1/26	Is Population Growth the Key?	Sims et al. Ch. 45 & 49 Wheeler Ch. 17	Lab 1 Report (Due before your next scheduled lab class)
	1/28	Sustainable Cities or Resilient Cities?	Hagen and Pijawka Ch. 8 and 10	
2	1/30	The Urban Footprint	Simkin et al. 2022. "Biodiversity Impacts and Conservation Implications of Urban Land Expansion Projected to 2050." *Proceedings of the National Academy of Science* 119, no. 12: e2117297119.	
	Week 4 2/2	The Urban *Ecological* Footprint		Reading Journal Discussion Forum 1 (Due at 11:59 p.m.)
	2/4	Life Cycle Thinking	Hagen and Pijawka Ch. 13	
3	2/6	Common Pool Resources	Wheeler Ch. 20, 21, & 23	Lab 2 Report (Due before your next scheduled lab class)

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	Week 5 2/9	The Paradox of the Growth Machine	Wheeler Ch. 18	Lab 3 Report (Due before your next scheduled lab class)
	2/11	The Possibilities of Governing for Urban Sustainability		
4	2/13	The New Regionalism	Wheeler pp. 198-202 Wheeler Ch. 22	
	Week 6 2/16	*President's Day (No class meeting)*		*No Lab Meeting.*
	2/18	Regional Governance		
	2/20	Industrial Ecology: The Science & Technology of Sustainability	Wheeler Ch. 8	
	Week 7 2/23	Urban Metabolism Concept and Models	Hagen and Pijawka Ch. 15 Wheeler Ch. 13 & 14	Story Map Proposals (Due before your scheduled lab class)
	2/25	Green Jobs and Eco-Industrial Parks	Wheeler Ch. 16	
5	2/27	The Nature of the City	Hagen and Pijawka Ch. 16 Wheeler Ch. 9 and pp. 312-322 Cronon, W. 1996. "The Trouble with Wilderness: Or, Getting Back to the Wrong Nature." *Environmental History* 1(1): 7-28.	
	Week 8 3/2	Restoring Urban Biodiversity	Cook, E. 2020. "Urban Ecology, Green Networks, and Ecological Design" pp. 205-222 in B. Hagen. and K.D. Pijawka (eds.) *Sustainability for the 21st Century* (3rd Edition). Dubuque, IA: Kendall Hunt.	Lab 4 Report (Due before your next scheduled lab class)
	3/4	Water Resources in Cities	Hagen and Pijawka Ch. 4, 11, and 12 Wolch et al. 2014. "Urban Greenspace, Public Health, and Environmental Justice: The Challenge of Making Cities 'Just Green Enough'" *Landscape and Urban Planning* 125: 234-244.	

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	3/6	History & Sustainability of the LA Rivier with a Virtual Tour (during regular class session at our regular classroom)		
	Week 9 3/9	Guest Lecture: Bridging Policy and Planning to Create Sustainable Urban Regions		*No Lab Meeting.*
	3/11	Midterm Review		
	3/13	Midterm Exam		
6	***Spring Recess***			
	Week 10 3/23	Mitigating GHG Emissions in Cities	Hagen and Pijawka Ch. 9 Wheeler Ch. 7	Story Map Outline (Due before your scheduled lab class)
	3/25	Adapting to Climate Change in Cities	Hagen and Pijawaka Ch. 16	
	3/27	Climate Change: ESEM and Agenda Setting		
	Week 11 3/30	Special Issue: Sustainable Transportation & Smart Growth	Hagen and Pijawka Ch. 14 Wheeler Ch. 10, 12, & pp. 291-312	Lab 5 Report (Due before your next scheduled lab class) Reading Journal Discussion Forum 2 (Due at 11:59 p.m.)
	4/1	Spatial Analysis for Sustainable Transportation		
7	4/3	History and EJ in the City	Hagen and Pijawka Ch. 7 Wheeler pp. 203-204	
	Week 12 4/6	Social and Spatial Analysis of Environmental "Bads"	Pulido, L., S. Sidawi, and R. O. Vos 1996. "An Archeology of Environmental Racism in Los Angeles," *Urban Geography* 17: 419-439.	Lab 6 Report (Due before your next scheduled lab class)
	4/8	Social and Spatial Analysis of Environmental Goods		

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8	4/10	The Problem of Consumption	Princen, T. 2002. "Consumption and its Externalities: Where Economy Meets Ecology pp. 23-42 in T. Princen, M. Maniates, and K. Conca (eds.) *Confronting Consumption*. Cambridge, MA: MIT Press	
	Week 13 4/13	Consumption in the Urban Landscape	Hagen and Pijawka Ch. 5 Wheeler Ch. 11	Story Map Progress Check-In (Due at the end of your lab session)
	4/15	Distancing of Production & Waste in a Global Economy	Sims et al. Ch. 24 & 44	
	4/17	Individual Versus Collective Responses to Consumption	Goldstein, N.J. and R.B Cialdini. 2008. "A Room with a Viewpoint: Using Social Norms to Motivate Conservation in Hotels." *Journal of Consumer Research* 35: 472-482. Maniates, M. 2002. "Individualization: Plant a Tree, Buy a Bike, Save the World?" pp. 43-66 in T. Princen, M. Maniates, and K. Conca (eds.) *Confronting Consumption*. Cambridge, MA: MIT Press.	
9	**Week 14** 4/20	Sustainability Indicators: Measurement and Reporting	McAslan, D. 2015. "Assessing Urban Sustainability: Using Indicators to Measure Progress" pp. 235-258 in K. D. Pijawka (ed.) *Sustainability for the 21st Century* (1st Edition). Dubuque, IA: Kendall Hunt. Wheeler Ch. 6	Story Map Final Presentation (Due as scheduled in person at your lab section meeting and submitted to the Lab Brightspace)
	4/22	Guest Lecture: Developing new Sustainability Sciences through GIS		
	4/24	Urban Models: How Can Geodesign Contribute?	Hagen & Pijawka Ch. 5 Guhathakurta, S. 2017. "Examining Urban Sustainability through Urban Models" pp. 111- 126 in B. Hagen and K. D. Pijawka (ed.) *Sustainability for the 21st Century* (1st Edition). Dubuque, IA: Kendall Hunt.	

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	Week 15 4/27	Third Places: Developing Community	TBA	Story Map Final Presentation (Due as scheduled in person at your lab section meeting and submitted to the Lab Brightspace)
	4/29	Sustainable Citizenship and Sustainable Livelihoods	Sims et al. Ch. 42 Bullen, A. and Whitehead, M. 2005. "Negotiating the Networks of Space, Time and Substance: A Geographical Perspective on the Sustainable Citizen." *Citizenship Studies* 9: 499-516.	
	5/1	Final Exam Review		
Final Exam – Wednesday, May 7th, 11:00 AM – 1:00 PM				

Laboratory Topics

Week 1 No Labs (Introductory Week)
Week 2 No Lab Meetings (Martin Luther King Birthday Holiday)
Week 3 Lab 1: Population Modeling for New York, NY
Week 4 Lab 2: Mapping the Urban Footprint of Atlanta, GA
Week 5 Lab 3: Economic Opportunity and Housing Affordability in Detroit, MI
Week 6 No Lab Meetings (President's Day)
Week 7 Story Map Introduction and Proposal Development
Week 8 Lab 4: Modeling Urban Forests Versus Income
Week 9 No Labs (Midterm Week)
Week 10 Story Map Working Session I
Week 11 Lab 5: Indexing Neighborhood Walkability
Week 12 Lab 6: Mapping Air Toxics
Week 13 Story Map Working Session II
Week 14 Story Map Final Presentations
Week 15 Story Map Final Presentations

Laboratory Protocol

Accessing Software for the Labs

Software to accomplish labs in our course will be provided via cloud-based internet services. It can be accessed from any standard desktop or laptop computer with a standard web browser and an internet connection. That said, it is not advisable to try to work from a phone or tablet computer. **If you have a laptop computer, please bring it along to the lab sessions. If you do not own a laptop computer, please borrow one from the Leavey Library.** If you use a Mac, please do not use Safari, but instead use Chrome or another standard browser to connect with Brightspace and software for labs.

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Course Content Labs

Some lab sessions are directly linked to the course lecture content (see titles above). After these lab sessions, you will work on self-guided work tasks using specialized geographic analysis tools and one or more geospatial datasets or computer modeling tools. Each of these 1-hour and 50-minute "content" lab sessions will provide a brief introduction to the lab, including when appropriate, demonstration and help with some key steps with the software. Following each content lab, the next week's lab session will also include a brief discussion of how the lab relates to key concepts of sustainable cities covered in the readings and lectures, and how these tasks might have been varied and/or enhanced if performed by professionals in a real-world setting.

If time remains in each lab session, students may choose to get started with the lab and ask questions as they come up. However, some the lab work may be conducted independently of the lab session. If you have work left after the lab session, you are advised to start early on it, so you have time to visit the instructor or a lab co-instructor in office hours if you run into difficulties. After completing the computer work, you will write a short lab report to submit to your lab co-instructor before the next lab session (see the specific deadline posted on the lab Brightspace submission link). Each of the 6 content lab reports are worth 3 points in the course.

Story Map Labs

The first story map lab session will introduce the story map process and software, and you will prepare a story map proposal to submit to your instructor. After this, you will work both inside and outside of the lab session time, using datasets and geospatial software to complete components of your story map. This will include connecting with data, analyzing data, and making maps.

For the lab classes marked "Story Map Working Session" above, each student will bring work to date to show your classmates and your lab co-instructor. After the first story map working session, you will submit a story map outline. The lab co-instructor will review your progress. The story map proposal and outline will be graded (for up to 3 points each) and students will receive 3 points credit/no credit for exhibiting a draft story map at the second story map working session. At one of the two final lab sessions, you will give a presentation of your story map. The oral presentation of the story map at the final lab session will be graded for each student for up to 3 points.

Statement on Academic Conduct and Support Systems

Academic Integrity

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the

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production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

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Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Culture Journey - *(213) 740-2101*

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

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Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

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